Merging Specialist Diffusion Models for Multi-Task Astronomical Image Restoration

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Abstract

Astronomical restoration pipelines chain several task-specific steps – star removal, defect correction and super-resolution among others – each usually handled by a dedicated tool. We ask whether such a pipeline can be collapsed into a *single* set of weights by merging specialist checkpoints post hoc, avoiding both multi-network serving and joint multi-task data. From a shared, purpose-pretrained initialization θ_0 we fine-tune three v-prediction DDPMs (one per task) and merge their task vectors with Task Arithmetic, TIES-Merging, and DARE-TIES. Since zero-shot merging suffers from destructive interference, we add a brief post-merge fine-tuning stage, tuned via a two-stage grid search over all seven degradation combinations on manually-curated Hubble Legacy Archive (HLA) imagery. Code is avaliable here.

1. Introduction

Restoring astronomical images often requires multiple sequential corrections, including star removal, detector defect restoration (dead/hot pixels), and super-resolution. A common deep learning approach is to train one specialist model per task, but this requires storing and executing multiple networks.

Model merging combines independently fine-tuned checkpoints into a single model without joint training. While effective for classification and language models, we find that it performs poorly when applied zero-shot to conditional diffusion models for low-level image restoration.

Our experiments indicate that zero-shot merging of diffusion specialists suffers from destructive interference that is not always reflected by pixel-wise metrics but becomes evident under perceptual evaluation. We further show that a short, balanced post-merge fine-tuning stage consistently improves the performance of Task Arithmetic, TIES, and DARE-TIES, suggesting that post-merge adaptation has a greater impact than the specific merging algorithm.

2. Related Work

Diffusion models. Denoising Diffusion Probabilistic Models (DDPMs) learn to invert a fixed noising process (Ho et al., 2020). We adopt the *v-prediction* parameterization (Salimans & Ho, 2022), a cosine noise schedule (Nichol & Dhariwal, 2021), and classifier-free-style condition dropout so that the network retains some unconditional generative capacity; the backbone is a conditional U-Net (Ronneberger et al., 2015) with self-attention (Vaswani et al., 2017) at low resolutions.

Model merging. Task Arithmetic (Ilharco et al., 2023) combines task vectors by adding scaled weight differences to a shared pretrained model. TIES-Merging (Yadav et al., 2023) reduces interference by trimming low-magnitude updates and resolving sign conflicts, while DARE (Yu et al., 2024) further sparsifies task vectors through random pruning and rescaling. All methods require the same pretrained origin, motivating our shared-θ_0 stage (Sec. 3.2).

These techniques were developed and are still predominantly evaluated on image classifiers and language models. Their use in diffusion has so far centred on *parameter-efficient* composition: merging lightweight LoRA adapters for concept- and style-level personalization – e.g. concept sliders (Gandikota et al., 2024), merging subject and style adapters (Shah et al., 2024), and decentralized multi-concept customization (Gu et al., 2023) – where each adapter edits a semantic direction rather than solving a different restoration problem. To our knowledge, coordinate-level merging (Task Arithmetic, TIES, DARE-TIES) of *fully* fine-tuned diffusion specialists that each solve a distinct low-level restoration task – and the observation that such merges require a post-merge repair stage to be usable – has not been investigated.

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3. Method

3.1. Dataset

Source imagery and pre-processing. We start from a manually curated set of color, already-drizzled HLA mosaics (Space Telescope Science Institute (STScI), 2006). Because exposure levels, noise floors, and dynamic ranges vary widely across targets, a single global stretch is inadequate; we instead apply an image-specific adaptive arcsinh stretch to each image before any degradation.

Dataset generation. For all three tasks the supervision target is always the fully clean, stretched image; only the *condition* (network input) differs. Each specialist therefore sees, as condition, an image degraded with only its own corruption, but is only ever supervised to recover the clean image – as opposed to θ_0's pretext task (Sec. 3.2), where the degradation is deliberately *not* tied to the task the source image came from.

For **star removal**, StarNetV2 (Misiura, 2020) is used only to *isolate* real stars (by subtraction from the clean image), not as ground truth. The extracted star patches are re-injected onto pristine starless backgrounds via linear blending, yielding controlled input–target pairs, thus avoiding distillation. For **dead/hot-pixel restoration**, defective pixels are flagged by a robust detector on the residual from a 5×5 median filter: a pixel is defective when its channel residual exceeds 6.0σ, with σ from the MAD scaled by 1.4826. As HLA composites combine independent filter exposures, defects are sparse and usually confined to one channel. Lastly, for **super-resolution**, each high-resolution image is degraded with Gaussian blur ($\sigma = 1.5$) and $\times 4$ cubic downsampling, then upsampled back to full resolution via bicubic interpolation to give a spatially aligned conditioning input for the DDPM.

After each of these processes is done, randomized patch sampling and data augmentation extract sub-images exclusively from regions verified by a geometric validity mask.

Lastly, to evaluate merged models under realistically compounded corruption, we build 7 sets by applying the corresponding combinations of single-task degradations *in sequence* to the same clean target.

3.2. Shared initialization θ_0

All three specialists are fine-tuned from a common checkpoint θ_0, obtained by pretraining the same architecture from a random initialization on a *generic conditional-restoration* pretext task, rather than starting each specialist from an independent random init, as required by the original merging methods themselves.

More specifically, θ_0 is trained to reconstruct clean images from synthetically degraded inputs. Degradations consist of 1–3 randomly selected operations from $\{$Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample-upsample, and small local occlusions$\}$, with independently sampled parameters and a 10% probability of a fully zeroed condition (classifier-free guidance dropout (Ho & Salimans, 2022)). This decorrelates image content from degradation type, preventing the pretrained model from learning task-specific biases.

3.3. Architecture

The same conditional U-Net (*ImprovedUNet*) is used, *strictly identical*, across θ_0 pre-training and all fine-tuning. It takes a 6-channel input (the noisy latent x_t concatenated with the conditioning map) through 4 down- and 4 up-sampling stages, using strided 4×4 convolutions and channel widths $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. Each stage is a residual *UNetBlock* (two 3×3 convolutions, GroupNorm with 4 groups, SiLU); a sinusoidal timestep embedding ($C = 64$) followed by a two-layer GELU MLP yields a 256-d vector added to every block. Multi-head self-attention (2 heads, LayerNorm) is applied at the deepest encoder stage, the bottleneck, and the first decoder stage, and a final convolution projects the 64-channel output to the 3-channel v-prediction.

3.4. Specialist fine-tuning and merging

Each specialist reuses θ_0's architecture and the v-prediction framework, incorporating a cosine noise schedule over $T = 200$ diffusion steps, an exponential moving average (EMA) with $\beta = 0.999$, and the AdamW optimizer. Fine-tuning is conducted exclusively on each specialist's respective (condition, target) pairs from Sec. 3.1. The optimization utilizes a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}, regulated by a cosine annealing learning rate scheduler over a total of 100 epochs.

Two-stage grid search. The scaling coefficients $\lambda = (\lambda_{\mathrm{SR}}, \lambda_{\mathrm{IR}}, \lambda_{\mathrm{SU}})$ are chosen by a two-stage search over a small Cartesian grid of per-task values. Every candidate merge is built, sampled with full DDIM, and scored by an averaged per-folder Z-score (LPIPS on any super-resolution folder, MSE otherwise; lower is better). *Stage one* ranks the entire grid on the three single-task folders (SR, IR, SU) and keeps only the top-K of combinations; *stage two* re-evaluates these survivors on the four compound-degradation folders and retains the best-scoring combination as the final merge. A fixed small subsample of each folder keeps the number of sampling passes affordable.

4. Results

4.1. Choosing the search objective

We first developed and validated the search objective on Task Arithmetic, iterating through several evaluation metrics; the best-performing configuration was then applied unchanged to TIES and DARE-TIES on the final iteration to maximise the results.

Merge selection. Initial model selection using pooled MSE consistently favored $\lambda_{\mathrm{SU}} = 0$, as the larger super-resolution error dominated the objective and effectively suppressed that task. Replacing pooled MSE with task-normalized MSE relative to each single-task expert yielded more balanced solutions. For combined-degradation datasets, where no specialist baseline exists, performance was normalized with respect to θ_0.

Perceptual Z-score. The final refinement addresses the mismatch between low MSE and visibly poor super-resolution output: on any folder involving super-resolution, LPIPS replaces MSE. To make heterogeneous per-folder scores comparable, each folder is standardised with a per-folder Z-score $z = (x - \mu)/\sigma$ computed across the candidate grid, and the resulting Z-scores are averaged to rank candidates.

However, when troubleshooting the winning Task-Arithmetic merge we found a striking disagreement between the two families of metrics. The merge attains the *lowest* MSE of the three yet by far the *worst* LPIPS: pixel error was rewarding a smoother but perceptually poorer prediction, and the gap survives much longer sampling, ruling out a sampling artifact. Although the task vectors are nearly orthogonal (cosine similarity $\approx$ 0.08–0.12), which normally favours task arithmetic, τ_{SU}'s much larger norm means even a small contribution from the other tasks perturbs its direction enough to visibly degrade quality.

Findings. Across the whole progression above, the evaluation metric mattered more than the search algorithm. Raw pooled MSE optimises total pixel error rather than balanced multi-task quality, and therefore repeatedly suppresses super-resolution; normalising against per-task baselines removes this pooling artifact and recovers a genuine three-task compromise. Only a perceptual objective, however, exposes the task-vector interference that pixel-wise error cannot see, so any merge of these experts should be validated perceptually before it is trusted.

4.2. Effect of post-merge fine-tuning

Applied zero-shot, all three merging methods produced visible artifacts and degraded reconstruction fidelity, confirming the destructive interference discussed above. We therefore fine-tuned each merged checkpoint with a short, deliberately conservative schedule that initializes *both* the network and its EMA copy from the merged weights and trains on a small, balanced mixture of all seven degradation combinations (SR, IR, SU and their pairings and triple). The diffusion formulation, optimizer and schedule are kept identical to the specialists (Sec. 3.4), so that only the initialization distinguishes this stage from single-task fine-tuning.

Findings. The effect is consistent across all three merging methods: the brief fine-tune resolves the interference introduced by merging and recovers multi-task quality that the zero-shot merges never reach. Quantitatively, the mean perceptual Z-score moves from worse- to better-than-average for every method (Table 1, Appendix), and the corresponding training curves (Fig. 2, Appendix) show a monotone decrease in both MSE and Z-score for train and validation, with the three methods converging to a comparable level.

	Task Arith.	TIES	DARE-TIES
Merged	+2.15	+2.05	+2.5
Merged + Fine-tuning	-1.15	-1.20	-0.80

Table 1. Mean perceptual Z-score (averaged over the seven evaluation folders; lower is better) before and after post-merge fine-tuning. The brief fine-tune moves every method from a positive (worse-than-average) to a negative (better-than-average) Z-score, and the three methods converge to a comparable level.

5. Discussion and Conclusions

We collapsed a three-stage restoration pipeline – star removal, dead/hot-pixel restoration and super-resolution – into a single set of weights by merging fully fine-tuned diffusion specialists sharing a pretrained initialization θ_0. Two findings stand out. First, evaluation is subtle: pixel-wise error cannot see the destructive interference between task vectors, which only a perceptual objective exposes (Sec. 4). Second, and more importantly, zero-shot merging alone is insufficient here: a short, balanced post-merge fine-tuning stage removes this interference and lifts every merged model's perceptual Z-score from worse- to better-than-average (Table 1).

Since Task Arithmetic, TIES and DARE-TIES converge to essentially the same quality once this stage is applied, the post-merge step – not the choice of merging algorithm – is what makes weight-space merging viable here. In practice, one can favour the simplest merging rule and spend the compute budget on a brief joint fine-tune rather than on an elaborate merging recipe.

6. Statement on the use of AI

Three AI assistants were used in distinct, clearly scoped roles:

- *Claude* assisted with code generation, architectural design decisions, debugging during training and sampling, and drafting this report.
- *Gemini* was used only for code generation.
- *ChatGPT* was used only for editing and polishing the writing of this report.

All architectural choices, training runs, and quantitative results reported here were produced, checked, and validated by the authors.

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Appendix

Implementation details

Degradation hyperparameters. Star patches are typically 64×64 px and re-injected by linear blending onto starless backgrounds. Dead/hot pixels are detected from the residual to a 5×5 median filter with a 6.0σ threshold, σ estimated from the MAD scaled by 1.4826; hot pixels approach the clipping value (1.0) and dead pixels drop toward the local noise floor (0.0). Super-resolution pairs use Gaussian blur ($\sigma = 1.5$), $\times 4$ cubic downsampling, and bicubic upsampling back to full resolution. The θ_0 pretext task applies 1–3 random operations from $\{$Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample–upsample, small local occlusions$\}$ with a 10% zeroed-condition dropout.

Training hyperparameters. All models use the v-prediction parameterization, a cosine schedule with $T = 200$ steps, EMA ($\beta = 0.999$), AdamW with learning rate 1×10^{-4} and weight decay 1×10^{-4}, and cosine annealing over 100 epochs. The U-Net widths are $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ with self-attention (2 heads) at the deepest encoder stage, the bottleneck, and the first decoder stage.

TIES merging steps. For completeness, TIES-Merging combines task vectors $\{\tau_i\}$ in three steps: *trim* each τ_i to its top-$k\%$ largest-magnitude coordinates (zeroing the rest); *elect* a per-coordinate sign by summing the trimmed vectors and taking the sign of the aggregate; and *merge* by averaging, per coordinate, only those entries whose sign agrees with the elected sign. DARE-TIES additionally applies random per-coordinate dropout with rescaling to each τ_i before this procedure.

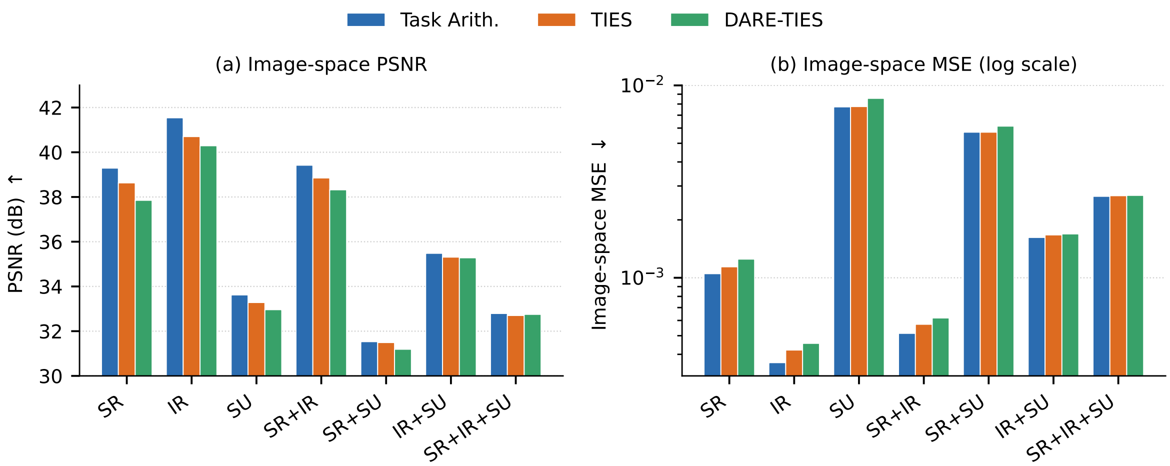

Figure 1. Per-folder comparison of the three fine-tuned merging methods on the image-space metrics: (a) PSNR (higher is better) and (b) MSE on a logarithmic scale (lower is better). The three methods track each other closely on most folders, with differences concentrated on the super-resolution-heavy sets.

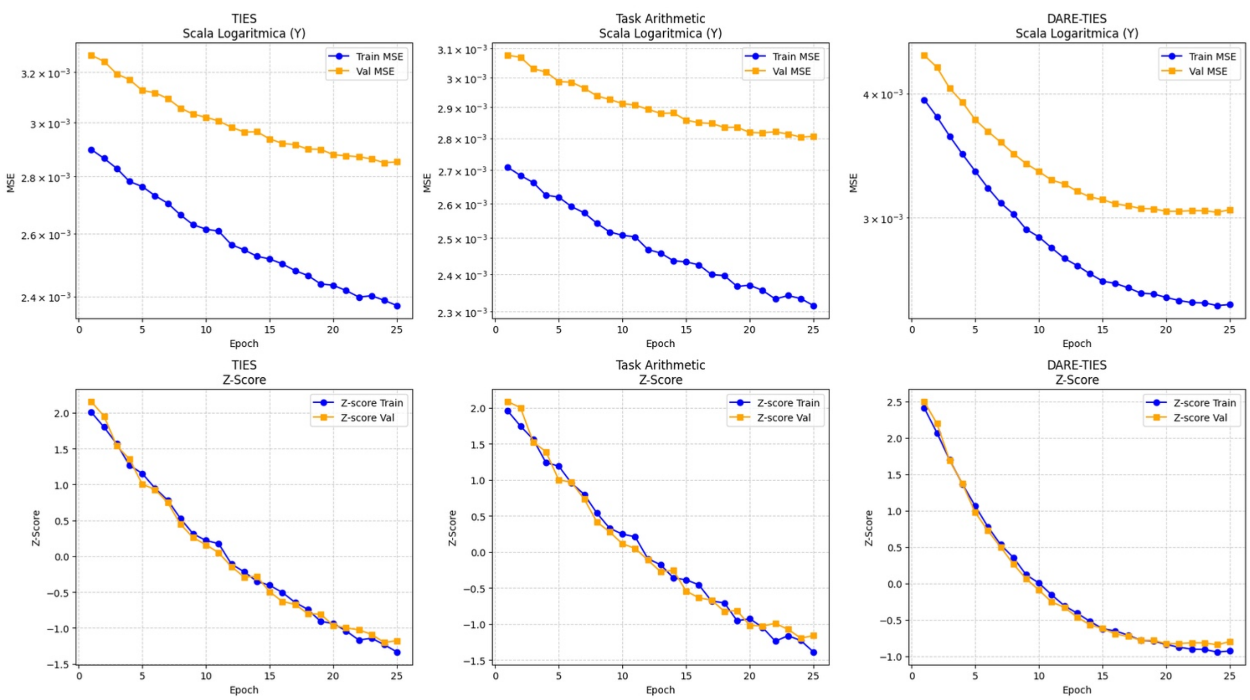

Figure 2. Post-merge fine-tuning progress over 25 epochs for TIES, Task Arithmetic and DARE-TIES. Top row: image-space MSE (logarithmic y-axis) for train and validation; bottom row: the averaged perceptual Z-score used as the selection objective. Every method shows a monotone decrease in MSE and a Z-score falling from positive (worse-than-average) to negative (better-than-average) values, consistent with Table 1.